Home Open Power

Q (1)

ISM Assignment -2

Azhar Ansari
2025ab05055
Section -4

Given data.

Population mean $\mu = 50cm$.

Population standard deviation $\sigma = 1cm$.

Sample size $n = 6$.

no of samples $= 10$.

(a) Sample mean of each sample.

$$\boxed{\bar{x} = \frac{1}{n}\sum x_i}$$

Sample-1, $\bar{x}_1 = \frac{49+50+51+50+49+51}{6} = 50$.

Sample-2, $\bar{x}_2 = \frac{50+51+49+50+50+50}{6} = 50$.

Sample-3, $\bar{x}_3 = \frac{51+50+49+50+51+49}{6} = 50$.

Sample-4, $\bar{x}_4 = \frac{48+49+50+50+51+52}{6} = 50$.

Sample-5, $\bar{x}_5 = \frac{50+50+51+49+50+50}{6} = 50$.

Sample-6, $\bar{x}_6 = \frac{49+50+50+51+50+50}{6} = 50$.

Sample-7, $\bar{x}_7 = \frac{50+49+51+50+49+51}{6} = 50$.

Sample-8, $\bar{x}_8 = \frac{51+50+50+49+50+50}{6} = 50$.

Sample-9, $\bar{x}_9 = \frac{49+50+51+50+51+49}{6} = 50$

Sample-10, $\bar{x}_{10} = \frac{50+50+49+51+50+50}{6} = 50$

Q1 (b)

mean & SD of sample distribⁿ

Azhar Ansari
2025a600055
Section 4

mean of sample mean $\mu_{\bar{x}} = \frac{1}{n}\sum \bar{x}_i$

$= \frac{50+50+50+50+50+50+50+50+50+50}{10} = 50.$

Standard deviation of sample means $(\sigma_{\bar{x}}) = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{6}} = 0.408.$

(c) probability that sample mean exceeds 50.5 cm.

$Z = \frac{50.5-50}{0.408} = 1.23$ $\quad P(\bar{x} > 50.5) = P(Z > 1.23)$
$= 0.1093$

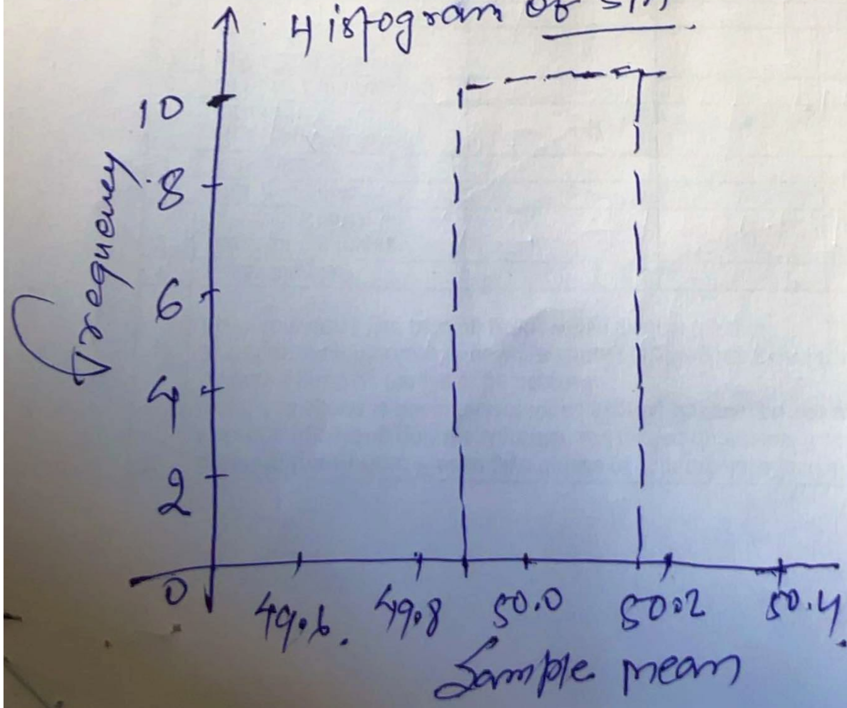
(d) probability that sample mean is below 49.5 cm

$Z = \frac{49.5-50}{0.408} = -1.23$. $\quad P(\bar{X} < 49.5) = 0.1093$.

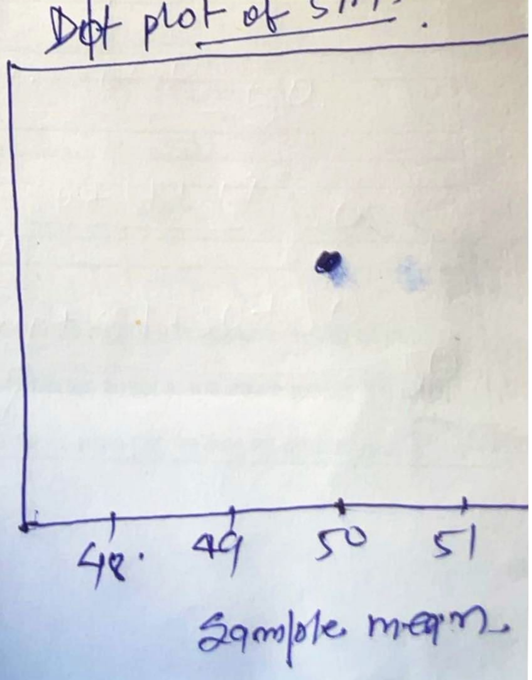
(e) → Both probabilities are low (~11%).
→ such extreme sample means are rare
→ This indicates process is stable and well controlled

(f) histogram and Dot plot $\bar{x}_i = [50, 50, 50, 50, 50, 50, 50, 50, 50, 50]$

Histogram of SM



Dot plot of SM.



Q1 (g) Effect of increasing sample size to 20

Azhar Ansari
20250505055
section-4.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{20}} = 0.223$$

- if I calculate z for 50.5 & 49.5

$$z = \frac{50.5-50}{0.223} = 2.24. \qquad z = \frac{49.5-50}{0.223} = -2.24$$

Conclusions:-

- increasing n reduces standard error
- Sampling distribution becomes narrower
- Estimates becomes more precise.

Q2. Given $n=24$, population mean $=50$.

(a) Sample mean and standard deviation

$$\bar{x} = \frac{\sum x_i}{N} = \frac{\begin{gathered}49+50+51+50+50+49+50+51+50 \\ +50+49+50+50+51+50+50+49+50+51 \\ +50+50+50+49+50\end{gathered}}{24}$$

$$\boxed{\bar{x} = 50.}$$

standard deviation.

$$s = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}}$$

$$s = \sqrt{\frac{\begin{gathered}(49-50)^2+(50-50)^2+(51-50)^2+(50-50)^2+(50-50)^2+(49-50)^2+(50-50)^2 \\ +(50-50)^2+(50-50)^2+(50-50)^2+(49-50)^2+(50-50)^2 \\ +(51-50)^2+\cancel{(51-50)^2}+(50-50)^2+(50-50)^2+(50-50)^2+(49-50)^2+(50-50)^2 \\ +(50-50)^2+(51-50)^2+(50-50)^2+(50-50)^2+(49-50)^2+(50-50)^2 \\ +(51-50)^2+(50-50)^2+(50-50)^2+(50-50)^2+(49-50)^2+(50-50)\end{gathered}}{24-1}}$$

$\simeq 0.79$

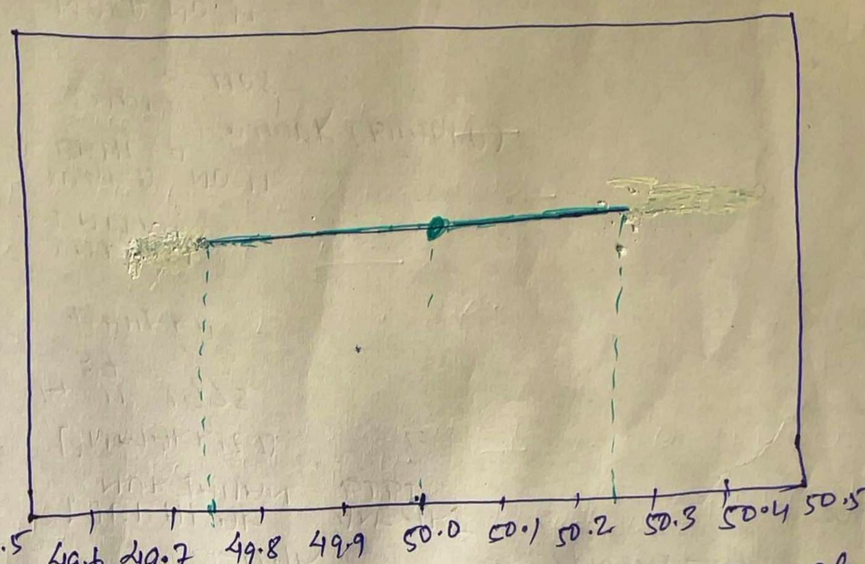
Q2 (b) 90% Confidence interval,

Azhar Ansari
20259b05055
S-4

$$Z_{0.05} = 1.645$$

$$CI = \bar{x} \pm z\frac{s}{\sqrt{n}} = 50 \pm 1.645\left(\frac{0.74}{\sqrt{24}}\right) = 50 \pm 0.25$$

$$\boxed{(49.75, 50.25)}$$



(c) Decision: Since 50 lies inside interval, Hence target mean is met.

(d) Width of CI: (i) increase sample size → CI narrows
(ii) increase confidence level → CI widens.

(e) If Lower limit <49.
→ inspect machinery
→ calibrate equipment
→ increase monitoring frequency

(f) If 99% CI & sample size.
→ Higher confidence → larger Z
→ to maintain same precision

$\boxed{n \propto Z^2}$ Sample size must increase

End